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Developmental toxicity potential of paclobutrazol in the rat

VERA LÚCIA DE CASTRO¹, KARINA P. GOES² and SELMA H. CHIORATO²

¹Embrapa Environment, Jaguariúna, SP, Brazil, ²School of Biology–PUCC, Campinas, SP, Brazil

Paclobutrazol is used as a fungicide and as a plant growth regulator, but there are few studies about its potential toxic effects. Experiments were conducted to determine whether this compound has adverse effects on the reproduction and development of offspring. The influence of 1.0 mg/kg paclobutrazol (10 × the acceptable daily intake–ADI), by oral exposure during gestational organogenesis of rats, was evaluated on maturational and behavioural aspects of offspring development. This dose did not promote evidence of maternal toxicity and the weight of gravid uterus, fetus, and ovary on days 16 and 20 of pregnancy were not affected. Also, the pesticide did not affect body weight gain of the dams and offspring. However, the pups' survival at weaning was impaired by paclobutrazol. Beyond that, study of the functional state of rat pups' nervous system at different stages of postnatal development revealed some differences in treated ones. Damage was observed in the expression of acoustic startle reflex and altered locomotion and/or rearing in an open-field apparatus in treated pups depending on their age. There were no observed alterations in swimming behaviour. The data support further studies of the potentially toxic effects that exposing mothers to this pesticide may have on their litters.

Introduction

The identification of environmental compounds that have adverse effects on reproductive health and animal development is particularly challenging. Toxic effects on newborns by environmental pollutants have been associated with maternal exposure to various pesticides, as developmental and reproductive effects, because of their employment worldwide (Barr *et al.* 1999). Protecting offspring from the developmental hazards of pesticides requires a comprehensive strategy capable of monitoring the patterns of pesticide use on a continuing basis (Landrigan *et al.* 1999). However, as the severity of man's impact on the environment worsened, it became clear that more sophisticated efforts were needed to systematically monitor the extent and effects of this input (Golden and Rattner 2003).

Paclobutrazol [(2RS, 3RS)-(4chlorofenil)-4,4-dimetil-2-(1H, 1,2,4-triazo-1-yl)pentam-3-ol] is used as a fungicide and plant growth regulator. It has been commonly used in agriculture and is frequently sprayed at the same time on various crops, especially fruits. It is frequently applied to mango trees 2–4 months before flowering which allows for a pre-harvest interval of 6–10 months. The acceptable daily intake (ADI) for paclobutrazol established with CODEX is

Correspondence: Vera Lúcia de Castro, Embrapa Environment, Rodovia SP 340, km 127,5 Jaguariúna, SP, 13820-000 Brazil. Tel.: 5519 3867-8739; Fax: 5519 3867-8740; E-mail: castro@cnpma.embrapa.br

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0.1 mg/kg body weight. The maximum residue limits (MRL) in fruits vary from 0.05 to 1.0 mg/kg, which would result in 0.5 mg/kg per apple.

Very little is known about the dissipation characteristics of this compound in the environment. Due to soil adsorption, paclobutrazol and its major decomposition products have low mobility in soils, since paclobutrazol is strongly bound to organic matter. The lack of polar functional groups in this molecule explains why it is absorbed into the hydrophobic sites on organic matter. Thus, causes of non-biological loss of paclobutrazol in soil are few since it is non-volatile and thermally stable (Jackson *et al.* 1996). The half-life of paclobutrazol in water was 24.4 days. Its solubility is 35 mg/l and the compound has high potential to leach into surface and ground water. This compound has the potential of maintaining submersed aquatic weeds at reduced heights while allowing them to retain useful functions such as oxygen production and fish habitation. In addition, the ability of natural microbial populations in small-scale aquatic sediments to degrade these compounds still needs to be investigated (Chand and Lembi, 1994).

Vergieva (1998) observed the exposure effects of 50 or 200 mg/kg paclobutrazol at 6–15 days of the pregnancy (organogenesis exposure) or a single dose of 200 or 500 mg/kg at 7, 9, 11 or 13 days of pregnancy on Wistar female rats. After a single treatment, a higher percentage of fetuses with external and visceral anomalies was observed. Some fetal malformations occurred with the single dose, mainly the largest dose. During the organogenesis exposure, there was loss of the maternal weight gain with both doses used. There was also a tendency for an increase in malformations at the largest dose, while no alterations were observed at the 50 mg/kg dose.

In view of these few results that studied higher dosages of paclobutrazol, an important issue regarding the toxicology of these compounds concerns these subtle effects on neurobehavioural development of mammalian progeny following *in utero* exposure to small dose levels; lower than those associated with maternal toxicity and more reliably to be expected in the environment. Thus, the present study was designed to analyse the influence of lower prenatal exposure to paclobutrazol on maturational and behavioural aspects of offspring development.

Vulnerability parameters can be used as tools to rank the severity of effects and the susceptibility of reproduction and developmental damage in a population upon exposure to a compound. The toxic effects of paclobutrazol have to be further studied since the knowledge about them is limited. In this way, some reproductive and developmental endpoints will be analysed in order to produce toxicity data of paclobutrazol in a challenging dose. The results obtained may aid further discussion about its possible toxic effects and safety.

Materials and methods

Animals and pesticide dosing

Animals Ninety-day-old Wistar male and nullipara female rats from our own colony, weighing 230 ± 15 g, were used. Males, previously determined to be fertile, were placed overnight with females. The presence of a vaginal plug or spermatozoa in vaginal smears of females was checked the following morning. If they were positive, plug day was considered gestational day zero. The animals were fed a commercially prepared laboratory diet (Purina Lab Chow). Pregnant females were placed in polycarbonate single cages with tap water and food available *ad libitum* and with hardwood chip bedding, at constant room temperature ($22 \pm 2^\circ\text{C}$) and humidity (70%). Animal rooms were maintained at a 12-hour light/dark cycle.

The day of parturition was defined as postnatal day 1 (PND 1). Within 24 h of delivery, all litters were counted, sexed, and weighed. The pups were separated by sex based on the anogenital distance. Litters with less than six pups were not included in the experiments. Efforts were made to retain equal numbers of males and females at a maximum of eight pups per litter during the first 2 days of life. The dams were checked twice daily (9:00 and 17:00 h) for parturition. Litters born after 17:00 h were considered as day 1 of life on the next day of observation.

Pups were weaned on PND 21 and housed by sex, five to a cage. Pup body weights in grams were recorded on PND 7, 14 and 21 for every pup from each litter. The pups' weights were considered as means for the litters. The litter was considered as the experimental unit in the physical landmarks evaluation, when all pups were tested. For behavioural studies, only male pups were tested. The reflex development was tested using three pups per litter identified with an ink spot. This protocol design was used in an attempt to minimize the influence on the obtained data of the effects of intrinsic values of each litter (Wise *et al.* 1995), such as maternal behaviour. Other pups from the same litter were kept for neuromotor development evaluation – open field and swimming tests (Zhu and Stadlin 2000). Only males were tested, since sex can influence the locomotion (Andrews 1996; Tropp and Markus 2001; Walker *et al.* 2001). In this case, the pups were randomly selected and marked with an ink spot. All litters were tested in a previously determined sequence.

Dosing Dams were randomly assigned to control or treatment groups, which received saline solution or technical grade paclobutrazol (Cultar[®], 250 g/l) respectively. Pesticides were suspended in 0.9% NaCl saline solution with the addition of two drops of Tween 80. Controls were injected only with saline plus Tween 80. The volume of Tween 80 was taken into account for the control or pesticide suspension final volume. Doses were expressed in mg/kg of the active ingredient (paclobutrazol) of the product. Considering that control and treatment dams were the same age, weighed approximately the same and were handled in the same way, the influence of these parameters should have similar impacts on offspring development and/or behaviour. Dams were injected with 1.0 mg/kg paclobutrazol or saline solution at 7, 11 and 15 days of pregnancy per gavage.

Reproductive performance The females had their vaginal smears examined daily for 12 days before mating and exposure to pesticides. The oestrous cycle phase of each one was observed and the number of each phase was compared between the groups by Student *t* test. This procedure was carried out to discard any female with problems on her vaginal oestrous cycle.

The females were divided into two groups: ten dams were exposed to paclobutrazol and seven were treated as controls. After mating, dams were weighed at 1, 7, 14 and 21 days of pregnancy and examined daily for signs of toxicity. After parturition, the neonates were examined for external malformations in their physical appearance. All were observed daily for mortality and physical signs of toxicity. The number of viable pups, at birth and on the 21st day of life, and also the fertility of the dams were determined. Viabilities at birth and weaning were calculated, respectively, as (a) the total number of stillbirths and the number of neonatal deaths within 24 h and (b) the total number of weaned pups and the number of pups born alive of the culled litter, and the results were expressed as a percentage. Fertility was defined as the ratio between pregnant females and the total number of females that mated. The results were also expressed as a percentage.

Another 40 females were divided into four equal groups ($n = 10$) exposed to paclobutrazol or treated as controls. They had their maternal uteri removed by abdominal section on day 16 or 20 of pregnancy to determine the gravid uterus, fetal, placental and ovary weight.

Landmark development All litters were observed using the following physical developmental parameters to evaluate the possible effects of pesticide exposure: pinna detachment (unfolding of external ear), development of fur, incisor teeth eruption, ear and eye opening and descent of testes (descending of testicles into the scrotum); with the days required for the appearance of these landmarks registered, until all pups in the litter were positive for the developmental parameter. Age at eye or ear opening was recorded when both eyes or ears were completely opened. Incisor eruption was examined daily to determine the first day of the bilateral appearance of the upper and lower incisors. The frequency of animals showing each of the above parameters and their time of appearance according to age were observed. As a basis of all postnatal behavioural studies, the appearance of physical signs had to be recorded and compared with the normal physical development of the strain and species used (Alder 1983).

All pups were removed from the nest with a paper towel, and placed on a heating pad to provide warmth before and after testing. The order of testing the litters was determined by chance.

Reflex development and neuromuscular maturation Throughout the lactation period, three male pups of each dam were evaluated for reflex development and neuromuscular maturation ($n = 30$ for paclobutrazol and $n = 21$ for control) to evaluate the possible effects of pesticide exposure. The animals showing each of the parameters tested were observed according to their age (Alder 1983).

The parameters tested were: (1) righting reflex (PND 2–5) – the newborn was placed on its back on a padded plain surface, and quickly set free. Time was measured until the animal had turned over on its ventral surface. Reflex was considered to be formed if the rat pups regained their position on four paws and the test was no longer than 30 s; (2) grip response (PND 3–7) – the pup was suspended with its forepaws on a horizontal rod and its grip strength was measured; (3) climbing time response (PND 6–11) – the pup's climbing time on an inclined screen (45°) made of a wire mesh was measured. The test was no longer than 120 s. This test reflects development of grasping reflex and grip strength; and (4) acoustic startle response (PND 11–14) – the presence or absence of a startle response that was elicited by a sharp noise which was produced near the pup was measured. The animal was put on a nonrestrictive padded plain surface. The test session began with a 5-min acclimatization period with a background noise. After that, rats were exposed to stimuli that consisted of one 80 dB pulse of a bell for 1 s.

Neuromotor development evaluation

Swimming test An aquarium ($25 \times 30 \times 60$ cm), filled with water (25°C) was used for the test between 10 a.m. to 12 noon. Three pups of each offspring were dropped from three times their own height into the centre of the aquarium ($n = 30$ for paclobutrazol and $n = 21$ for control) on PND 7, 14 and 21. They were observed for 10 s. These rats were used haphazardly.

Swimming performance was evaluated according to the position of the nose and head on the surface of the water to measure the motor development (St. Omer and Mohammad, 1987). After the test, the animals were dried and returned to the home cage. The angle of swimming

was rated as follows: 0—nose below surface; 1—nose at or above surface; 2—nose and top of head at or above surface but ears still below surface; 3—same as 2 except water line at mid ear level; 4—same as 3 except water line at bottom of ears.

Open-field studies The activity behaviour of male offspring on PND 21, 30 and 60 was evaluated in an open-field arena (Castro *et al.* 1992). The same two pups per offspring in each one of the experimental groups were used for each day of the test ($n = 20$ for paclobutrazol and $n = 14$ for control).

Hand-operated counters and stopwatches were used to score ambulating frequency (number of units entered), rearing frequency (number of times the animal stood on hind legs) and immobility time (total time in seconds without any spontaneous movements). The experiments were blind to the treatment group.

For this study, each animal was placed in the centre of the arena and the behavioural parameters were assessed for 6 min, divided into two equal periods. After each animal observation, the arena was washed with a 5% alcohol-water solution. To minimize possible circadian influences on behaviour, experimental and control animals were observed intermixed.

Statistical procedures The variables of percentage in the reproductive performance were analysed by χ^2 test, and to determine the statistical significance of body gain weight among groups, analysis of variance (ANOVA) for profile data (Morrison 1976) using the procedure GLM of SAS (1988) was used. Physical landmarks and reflex development parameters were analysed using one-way ANOVA and the data are presented as means ($\pm$ SEM). The changes in body weight among groups and open-field variables were evaluated by repeated ANOVA using the procedure GLM of SAS, since the same animals were analysed at various ages.

For the analysis of the pups' survival, the pups' days of life and a χ^2 test analysing proportions of deaths were utilized. The variables of reproductive performance, gravid uterus, fetal, placental and ovary weight were analysed via Student *t*-test. The scores of the swimming test were compared by ANOVA to determine statistical significance. The level used to establish statistical significance was $P < 0.05$.

The personnel that monitored the tests were blind to the treatment group.

Results

Reproductive performance and viability of pups

The fertility of the dams as well as the number of viable pups at birth were not affected by treatment. However, the end of the nursing period showed a significant decrease in survival of experimental pups (21st day of life— $\chi^2 = 4.45$; log-rank = 4.25; $P = 0.03$) (Table 1). The majority of deaths occurred on PND 1. The last day for this finding was PND 18 in the experimental group and PND 2 in the control group. The experimental group presented 17% of deaths (17/81) and the control group 4% (2/43). The reason for this remains unclear since the neonates did not show any physical signs of toxicity and there were no statistical differences in body weights ($F = 1.30$; $GL = 1.15$; $P = 0.27$) of control and treated animals on PND 1 (5.98 ± 1.16 and 5.53 ± 1.09); PND 7 (12.60 ± 1.84 and 11.66 ± 1.25); PND 14 (24.01 ± 4.34 and 20.53 ± 2.23) and PND 21 (37.18 ± 8.93 and 36.52 ± 5.07), respectively. Also, the maternal weight gain was similar in control and treated groups tested (232.71 ± 21.00 and 253.00 ± 32.80 at the beginning of pregnancy and 266.00 ± 20.36 and 283.30 ± 33.70 at day 21 of pregnancy respectively— $F = 2.30$; $GL = 1.15$; $P = 0.15$).

Table 1. Viability of pups after maternal exposure to paclobutrazol ($n = 10$) or to control solution ($n = 7$), v.o. during pregnancy. The number of viable pups at birth and the end of lactation on the 21st day of life was determined (χ^2 test, $^aP < 0.05$). Data were expressed as percentage of each parameter observed

Parameters	Groups	
	Control	Paclobutrazol
Fertility of dams	100.0 (7/7)	100.0 (10/10)
Viability of pups at birth	100.0 (45/45)	100.0 (98/98)
Viability of pups at weaning	95.55 (43/45)	82.65 (81/98) ^a

Table 2. Summary of pregnancy data on days 16 or 20 of pregnancy of females exposed to paclobutrazol ($n = 10$) or to control solution ($n = 7$), v.o. during pregnancy ($n = 10$). Data were expressed as mean $\pm$ sd (grams), Student t -test, $^aP < 0.05$

Weight (g)	Groups			
	Paclobutrazol		Control	
Days of pregnancy	16	20	16	20
Gravid uterus	10.41 $\pm$ 5.67	32.10 $\pm$ 11.96	12.56 $\pm$ 3.31	33.56 $\pm$ 5.63
Fetus	0.31 $\pm$ 0.15	2.20 $\pm$ 0.28	0.24 $\pm$ 0.07	2.04 $\pm$ 0.18
Placenta	0.24 $\pm$ 0.10	0.54 $\pm$ 0.10 ^a	0.20 $\pm$ 0.02	0.46 $\pm$ 0.05
Ovary	0.16 $\pm$ 0.03	0.19 $\pm$ 0.02	0.17 $\pm$ 0.02	0.21 $\pm$ 0.06

The weight of gravid uterus, fetus, and ovary were not affected on days 16 and 20 of pregnancy by treatment. However, the placenta weight was enhanced on day 20 but not at day 16 of pregnancy (Table 2).

Landmark development, reflex development and neuromuscular maturation

All litters were observed daily to evaluate the possible effects of pesticide exposure. The results were analysed by ANOVA to verify if there was significant slowing or speeding up of developmental stages or one of the physical developmental parameters.

The animals, regardless of treatment, were able to reach each of the physical parameter's frequencies of the tested landmarks according to the pups' age. The frequency of animals showing each of the parameters tested on a certain day of life was not different in both groups.

The appearance of physical parameters to all control and treated pups occurred respectively as follows: pinna detachment PND 2.71 $\pm$ 0.48 and 3.10 $\pm$ 0.73; development of fur PND 5.40 $\pm$ 0.37 and 5.14 $\pm$ 0.51; incisor teeth eruption PND 4.57 $\pm$ 0.78 and 4.90 $\pm$ 0.73. The appearance of ear and eye opening occurred respectively on PND 11.57 $\pm$ 0.97 and 11.90 $\pm$ 1.19, and 14.00 $\pm$ 0.57 and 14.80 $\pm$ 1.68. The physical parameters of sexual maturing (descent of testes into the scrotum) occurred on PND 23.14 $\pm$ 1.95 and 23.70 $\pm$ 1.70 respectively to control and treated pups.

Three male pups of each dam were evaluated for their reflex development and neuromuscular maturation. The parameters tested were: righting reflex; grip response; climbing time response and acoustic startle response. The study of functional state of the rat pups' nervous system at different stages of postnatal development revealed differences as to when the reflex was expressed. The appearance of acoustic startle response was delayed for paclobutrazol rat pups in relation to control ones ($\chi^2 = 4.76$; GL = 1; $P < 0.03$) (Table 3).

Swimming test

The swimming performance was evaluated according to the position of the nose and head on the surface of the water. The pups' performance observed was similar in all groups tested ($F = 0.76$; GL = 1.15, $P = 0.39$). The swimming behaviour medium scores of offspring exposed to paclobutrazol were 1, 2, 4 and to control solution was 0, 2, 4 on PND 7, 14 and 21 respectively.

Open-field studies

The activity behaviour of male offspring on PND 21, 30 and 60 was presented in Table 4. Open-field testing resulted in an alteration of the locomotion on PND 21 and rearing frequency in the experimental group in relation to the control group at all days of open-field test when ANOVA was made ($F = 11.16$; GL = 1; 16; $P < 0.004$ and $F = 6.41$; GL = 1; 80; $P < 0.01$ respectively).

Discussion

The traditional approaches to testing for the reproductive and developmental effects of environmental agents like pesticides have been fairly standard for a number of years. Within the last decade, a number of efforts have been undertaken to improve the testing protocols

Table 3. Performance of pups exposed to paclobutrazol or to control solution, v.o., during pregnancy. Three pups of each litter were used for the performance evaluation ($n = 30$ and 21 respectively). Data were expressed as mean $\pm$ SD to show the parameter tested, ANOVA, ^a $P < 0.05$

<i>Landmark development group</i>	<i>Days of life to show the developmental landmark</i>
Righting reflex	
Control	2.95 $\pm$ 0.97
Paclobutrazol	2.90 $\pm$ 0.96
Grip response	
Control	3.61 $\pm$ 0.74
Paclobutrazol	3.60 $\pm$ 0.72
Climbing time response	
Control	7.47 $\pm$ 1.83
Paclobutrazol	7.83 $\pm$ 1.53
Acoustic startle response	
Control	12.23 $\pm$ 0.94
Paclobutrazol	13.20 $\pm$ 1.66 ^a

Table 4. Open-field performance of offspring exposed to paclobutrazol ($n = 10$) or to control solution ($n = 7$), v.o., during pregnancy. The parameters observed were locomotor (LO) and rearing (RE) frequencies and immobility time (ID). Data were expressed as mean $\pm$ SD and were analysed by ANOVA followed by Fisher test—^a $P < 0.05$

Parameter and day of observation	Groups	
	Control	Paclobutrazol
LO 21	7.57 $\pm$ 4.88	22.65 $\pm$ 16.11 ^a
LO 30	26.43 $\pm$ 10.89	24.00 $\pm$ 17.65
LO 60	16.14 $\pm$ 13.55	17.25 $\pm$ 12.76
RE 21	7.07 $\pm$ 3.81	12.65 $\pm$ 5.72 ^a
RE 30	12.21 $\pm$ 5.13	14.90 $\pm$ 11.07 ^a
RE 60	5.14 $\pm$ 4.59	5.80 $\pm$ 3.30 ^a
ID 21	4.07 $\pm$ 9.27	3.90 $\pm$ 6.77
ID 30	10.36 $\pm$ 21.52	18.25 $\pm$ 29.12
ID 60	17.50 $\pm$ 22.93	17.75 $\pm$ 38.34

available and to add endpoints that provide a better characterization of these effects of chemical exposure.

The paclobutrazol did not show damage of the developmental landmarks at the dose tested. The effects observed in the present work could mainly be a consequence of prenatal exposure since maternal treatment was restricted to a phase of the gestation period and, because of this, it might be assumed that there were no circulating levels of the pesticides at birth.

The pesticide did not appear to cause congenital malformations at term. Besides, the placentas' weight enhancement on day 20 of pregnancy may be an artifact; since no other significant effects were observable in pregnancy data during the tests.

However, pups from dams exposed to paclobutrazol during their gestational period, presented a diminution of survival at the end of lactation. There were no differences in body weight of the pups, yet 17% of the pups in the treated group died before the end of weaning. This effect is unlikely to be a consequence of a general toxic effect of the treatment because the dose of paclobutrazol given to pregnant rats did not cause overt signs of maternal toxicity and did not affect maternal weight.

The ways to explain the alterations of survival caused by these pesticides are devisable, but to further understand these effects on rat offspring, additional reproductive and toxic-kinetic studies should be carried out to determine the extent of exposure to paclobutrazol *in utero*. This effect may be due to the interaction of pesticide with some physiological system that is determinant of somatic, physical and behavioural development in postnatal life. The present data are a confirmation of the sensitivity of the animals, in the perinatal period, to pesticides whose effects are not always easily predictable.

The acoustic startle reflex is a relatively simple protective behaviour, which is mediated by a well-known oligosynaptic neuronal circuit. The use of a relatively simple startle reflex could demonstrate several important forms of behavioural plasticity including habituation, fear potentiation and sensorimotor gating. Although the acoustic startle response is a simple behaviour, the regulating mechanism is complex (Zhu and Stadlin 2000). Several brain areas

with different critical development periods such as the subcortex, cerebellum and labyrinthine system are involved in these tests.

Locomotor activity is often used as an index of an animal's reactivity and adjustment to novel and, to some extent, stressful situations. Therefore, xenobiotic maternal exposure may alter the manner in which the progeny react and adapt to novel test environments as evidenced by the open field test (Kishi *et al.* 1995). It seems important to evaluate the present alterations with more details, since they may be related to the physiological systems also involved in acoustic startle response pesticide effects.

Alterations in nervous system development and its system plasticity, such as alterations in cholinergic function, which is implicated in cognitive functions (Everitt and Robbins 1997; Winkler *et al.* 2000), can be evidenced by this test. Furthermore, dopaminergic function is correlated with motor behaviour (Castro *et al.* 1992), and the open-field test can also reveal some disturbance at this neuro-transmitter system.

Alterations in the locomotor (PND 21) and rearing frequencies (PND 21, 30, 60) were observed at the open-field apparatus. However, the data showed large standard deviations were obtained and the immobility time findings of this study should be regarded with care. The early experiences of the young pups during the performance of many tests may itself be a stressful procedure. As a consequence, pups' responses may become erratic, regardless of which of the testing systems is applied. The early neurochemical effects on the brain can have long-term behavioural consequences for the developing organism (Bekkedal *et al.* 1999).

Although the open-field test could reveal alterations at exploration and motor activity (Castro *et al.* 1992), the protocol design of this test should be improved in future studies, to produce more uniform results. Depending on the experimental design, where the duration of testing is short for example, results may be biased (Hall *et al.* 2000). Since environmental manipulations may interact with the open-field results and processes like emotionality, exploration, learning and memory, and anxiety among others (Lehmann *et al.* 2000; Zimmerman *et al.* 2001), a variety of contexts will be employed to most sensitive open-field tests in order to differentiate the various behaviour expressions involved in it. In this way, more sensitive endpoints would have perhaps picked up different effects at low dose levels of pesticides. In spite of this, the findings of this study should be regarded with care because the activity of the pups should have been sufficient to detect any delayed significant negative impact on motor activity resulting from that exposure.

Conclusion

The results showed that when paclobutrazol at $10 \times$ ADI was administered to rats, few changes were observed. At this point there is no evidence that ADI is associated with a risk to human and/or environmental health. The characterization of these few affected endpoints by prenatal exposure to paclobutrazol is an indication toward a more specific analysis of maturational and behavioural effects. The data to be obtained may be useful to better explain the present results.

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